

Certus Text Command Reference

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Contents

About This Manual	5
Audience	5
Conventions Used in This Manual	5
Related Documents	6
1	8
Global Configuration Variables	8
eco_certus_client_install_dir	8
2	10
Manual ECO Commands	10
replay_distributed_eco	10
3	13
Signoff ECO Commands	13
eco_opt_design	14
distribute_certus_command	19
export_distributed_eco	27
get_design_mode	29
generate_summary	30
implement_distributed_eco	42
set_eco_opt_mode	44
set_design_mode	49
start_distributed_eco_clients	52
time_design_signoff	53
4	58
System Commands - Basic Tool-Tcl Commands	58
start_gui	58

About This Manual

The *Certus Text Command Reference* manual describes the syntax for the Certus Closure Solution text commands.

Audience

This manual is written for experienced designers of digital integrated circuits. Such designers must be familiar with design planning, placement and routing, block implementation, chip assembly, and design verification. Designers must also have a solid understanding of UNIX and Tcl/Tk programming.

Conventions Used in This Manual

This section describes the typographic and syntax conventions used in this manual.

<i>text</i>	Indicates text that you must type exactly as shown. For example: <code>report_annotated_check -missing_setup</code>
<i>text</i>	Indicates information for which you must substitute a name or value. In the following example, you must substitute the name of a specific file for <i>worst_entries</i> : <code>report_clock_timing -type skew -nworst worst_entries</code>
<i>text</i>	Indicates the following: • Text found in the graphical user interface (GUI), including form names, button labels, and field names • Terms that are new to the manual, are the subject of discussion, or need special emphasis • Titles of manuals

[]	Indicates optional arguments. In the following example, you can specify none, one, or both of the bracketed arguments: <pre>command [-arg1] [arg2 value]</pre>
[]	Indicates an optional choice from a mutually exclusive list. In the following example, you can specify any of the arguments or none of the arguments, but you cannot specify more than one: <pre>command [arg1 arg2 arg3 arg4]</pre>
{ }	Indicates a required choice from a mutually exclusive list. In the following example, you must specify one, and only one, of the arguments: <pre>command {arg1 arg2 arg3}</pre>
{ [] [] }	Indicates a required choice of one or more items in a list. In the following example, you must choose one argument from the list, but you can choose more than one: <pre>command {[arg1] [arg2] [arg3]}</pre>
{ }	Indicates curly braces that must be entered with the command syntax. In the following example, you must type the curly braces: <pre>command arg1 {x y}</pre>
...	Indicates that you can repeat the previous argument.
.	Indicates an omission in an example of computer output or input.

Related Documents

For more information about Certus, see the following documents. You can access these and other Cadence documents using Doc Assistant (online documentation system).

- [New Features in Certus](#)

Provides information about new features in the release of Certus.

- [*Certus User Guide*](#)
Provides information about using various Certus features.
- README file
Contains installation, compatibility, and other prerequisite information for this release. You can read this file online at downloads.cadence.com.

Global Configuration Variables

eco_certus_client_install_dir

eco_certus_client_install_dir

Default: " "

Specifies an Innovus version for the Certus client. The Certus client installation directory specified by this TCL variable overwrites any other path specified for the Certus client using any other mechanism, including \$PATH.

You need to provide the path to the <install_dir>/lnx86 directory of the Certus client (Innovus) installation.

To set this global variable, use the `set` command.

Note: This global variable must be specified before `set_eco_opt_mode -distributed_opt_block_restore_file`.

Examples

- The following command allows you to point to the Certus client installation directory:

```
certus> set eco_certus_client_install_dir  
/icd/flow/INNOVUS/INNOVUS231/latest.isr1/lnx86
```

- The following command provides a list of pointers to the Innovus database that can be loaded on Certus clients:

```
certus> set_eco_opt_mode -distributed_opt_block_restore_file blkRestore.txt
```

- The following command restores all design views from the `saveDir` directory, and generates view-specific reports in directories with the corresponding view name within the `pdfDir` directory:

```
certus> distribute_read_design -restore_design all -restore_dir saveDir -outdir  
pdfDir
```

Manual ECO Commands

replay_distributed_eco

```
replay_distributed_eco  
[-help]  
-block_eco_file_list <string>
```

Supports the Non-Interactive Manual ECO mode. This command sources the Innovus-compliant ECO files that are specified as input using `block_eco_file_list <file>`. The command specified in these ECO files will first be executed on the respective Certus clients before being processed for execution on the manager. After this, the processed commands will be executed (block by block) on both the manager and the Tempus clients.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>replay_distributed_eco</code> parameter. For a detailed description of the command and all its parameters, use the man command: <code>man replay_distributed_eco</code>.
<code>-block_eco_file_list</code> <code><string></code>	Specifies the list of Certus client block module names and the corresponding ECO files as input. You can also specify multiple ECO files for the same module. Note: You must provide the input files in the sequence of the optimization calls. The format of the file is presented below: <pre><block-module-1> <path-to-block-1-ECO-file> <block-module-2> <path-to-block-2-ECO-file> ... <block-module-N> <path-to-block-N-ECO-file></pre>

Examples

- The following command refers to a file (`blockecofile`) pointing to all Innovus-compliant Manual ECO TCL files that are to be replayed :

```
certus> replay_distributed_eco -block_eco_file_list blockecofile
```

Below is a template for `blockecofile`:

```
blockName1 blockName1_innovus.tcl
blockName2 blockName2_innovus.tcl
```

Refer to the [Sample Template Script](#) section of the *Certus User Guide*.

Signoff ECO Commands

The list of Signoff ECO commands that can be specified at the Certus command prompt is provided below:

- `eco_opt_design`
- `distribute_certus_command`
- `export_distributed_eco`
- `get_design_mode`
- `generate_summary`
- `implement_distributed_eco`
- `set_eco_opt_mode`
- `set_design_mode`
- `start_distributed_eco_clients`
- `time_design_signoff`

eco_opt_design

```
eco_opt_design  
[-help]  
[-all]  
[-area]  
[-drv]  
[-dynamic]  
[-hold]  
[-leakage]  
[-power]  
[-report_dir]  
[-report_full_clock_path]  
[-setup]  
[-update_timing]
```

Lets you choose between the timing closure flow or the timing and power closure flow in Certus. In addition, it honors the signoff timing from Tempus clients, and performs the physical-aware distributed optimization through each Certus client.

Note: This command requires a 64-bit executable.

Parameters

-help	Outputs a brief description that includes type and default information for each <code>eco_opt_design</code> parameter. For a detailed description of the command and all its parameters, use the man command: <code>man eco_opt_design</code>.
-all	Performs DRV, Setup, Hold, and Power optimizations in the Path-based Analysis (PBA) mode. Also, this parameter enables ECO routing and parasitic extraction using the <code>implement_distributed_eco</code> command and provides signoff timing analysis using the <code>time_design_signoff</code> command. Note: Aggressive Setup optimization at the

	expense of Hold is supported if you use the <code>set_design_mode -flowEffort extreme</code> setting. • Performs power closure along with timing closure if <code>set_design_mode -powerEffort low</code> or <code>high</code> setting is used. • Performs aggressive power optimization if you use the <code>set_design_mode -powerEffort high</code> setting. • Performs only timing closure (DRV/Setup/Hold) as a default flow. • Prints a detailed timing report using the <code>time_design_signoff</code> command, which includes the full clock path when the <code>-report_full_clock_path</code> parameter is specified along with <code>eco_opt_design -all</code>. You can specify the directory to save these reports using the <code>-report_dir</code> parameter. • Prints a detailed power summary report using the <code>time_design_signoff -report_power</code> command. • This parameter does not replace all customized fine-tuned flows. It provides the Certus users with easy access to Timing and Power ECO closure technology. However, the existing Certus flow using individual optimization steps within Certus is still supported.
<code>-area</code>	Reclaims area on the netlist without causing any DRV/Setup/Hold degradation.

-drv	Optimizes DRV violations. DRV optimization fixes violations in the following order: • glitch • drv (max_transition/max_capacitance violations) • xtalk delta delay Note: • DRV optimization supports buffer addition and resizing. • CellEM violations are fixed only when <code>set_eco_opt_mode -fix_cell_em</code> is set to <code>true</code>.
-dynamic	Prints the initial and final power numbers. It optimizes the dynamic power to reduce the total power consumption.
-hold	Optimizes the hold violations.
-leakage	Optimizes leakage power. Leakage optimization supports only VT swapping. Note: To optimize leakage using resizing in addition to VT swapping, use the following: <pre>set_eco_opt_mode -power_opt_focus eco_opt_design -power</pre>
-power	Optimizes total power in the design. During total power optimization, the software performs swapping and sizing on both the combinational and sequential cells, and also does buffer removal when enabled. It reduces leakage, internal, and switching power in the design.
-report_dir string	Specifies the directory to save reports.

<code>-report_full_clock_path</code>	Generates a detailed timing report that contains the full clock path.
<code>-setup</code>	Optimizes the setup timing violations. The setup optimization supports buffer addition, VT swapping, and resizing.
<code>-update_timing</code>	Generates updated timing data for use during optimization.

Examples

- The following commands perform timing closure (DRV, Setup, Hold) in the PBA mode:

```
certus> set_design_mode -powerEffort none -flowEffort standard
certus> eco_opt_design -all
```
- The following commands perform timing and power closure (DRV, Setup, Hold, and Power) in PBA mode:

```
certus> set_design_mode -powerEffort low|high -flowEffort standard
certus> eco_opt_design -all
```
- The following commands perform timing and power closure in PBA mode, and generates the full clock path report in the directory specified by the user with `-report_dir`:

```
certus> set_design_mode -powerEffort low|high -flowEffort standard
certus> eco_opt_design -all -report_full_clock_path -report_dir <>
```
- The following commands perform timing closure with aggressive Setup fixing at the expense of Hold timing in PBA mode:

```
certus> set_design_mode -powerEffort none -flowEffort extreme
certus> eco_opt_design -all
```
- The following command performs Hold fixing while not degrading the Setup timing:

```
certus> eco_opt_design -hold
```
- The following command recovers area by performing cell downsizing and buffer deletion while ensuring not to create any new Setup/Hold/DRV violations.

```
certus> eco_opt_design -area
```

Related Commands

- `implement_distributed_eco`
- `set_eco_opt_mode`
- `time_design_signoff`

distribute_certus_command

```
distribute_certus_command
[-help]
<command> [-designs <string>]
```

Queries the physical attributes or properties from Certus clients or verifies the placement status of a few instances in the design. The `distribute_certus_command` command can be run at any point after `start_distributed_eco_clients` is specified. The tool waits for the design to be loaded on the Certus clients before running the commands that are passed through `distribute_certus_command`.

Note:

The commands passed to `distribute_certus_command` should not impact routing, change the `don't_touch/don't_use` attribute, or trigger a new parasitic extraction. In addition, the tool should not reset any parasitics or generate any timing reports. These are not included because they may desynchronize the Certus client manager from the Certus clients.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>distribute_certus_command</code> parameter. For a detailed description of the command and all its parameters, use the man command: <code>man distribute_certus_command</code> .
<code><command></code>	Specifies the command for which you want to query the physical attributes/properties from Certus clients.
<code>-designs</code> <code><string></code>	Specifies the list of partitions for which <code>distribute_certus_command</code> is executed.

Examples

- The following command queries `report_annotated_parasitics` from each partition and sends the output to the Certus clients. The output shows a summary report, which is a [list of annotated and unannotated nets](#):

```
certus> distribute_certus_command {report_annotated_parasitics}
```

The following output is generated:

```
*info: output from client dtmf_recvr_core(0).

=====

#####

# report_annotated_parasitics: Thu Feb 20 01:36:20 2025

#   view: setup_func

#   -list_not_annotated

#####

#

# list syntax: type: net_name [nrCap Cap nrXCap XCap nrRes Res]

# unit: pF, Ohm

# Summary of Annotated Parasitics:

+-----+
| Net Type          | Count | Annotated (%) | Not Annotated (%) |
+-----+-----+-----+-----+
| total              | 1010 | 881 87.23%   | 129 12.77%       |
+-----+-----+-----+-----+
| assign (*)         | 3    | 0 0.00%     | 3 100.00%        |
| 0-term:floating (*) | 23   | 0 0.00%     | 23 100.00%       |
| no drive (*)       | 33   | 0 0.00%     | 33 100.00%       |
| 1-term:no load     | 57   | 0 0.00%     | 57 100.00%       |
| real net (complete) | 861  | 848 98.49%  | 13 1.51%         |
| real net (broken)   | 33   | 33 100.00%  | 0 0.00%          |
| zero capacitance net | 0    | 0 0.00%     | 0 0.00%          |
+-----+-----+-----+-----+
```

Note for Net Types marked "(*)": Such nets are never timed, but reported here for informational purpose.

+-----+				
Annotated	Res (KOhm)		Cap (pF)	XCap (pF)
+-----+				
Count	10514		10782	2480
Value	59.9570		4.1717	0.8583
+-----+				

*info: output from client ptn_wrapper(1).

=====

```
#####
# report_annotated_parasitics: Thu Feb 20 01:36:20 2025
#   view: ptn_wrapper_1_setup_func
#   -list_not_annotated
#####
#
# list syntax: type: net_name [nrCap Cap nrXCap XCap nrRes Res]
# unit: pF, Ohm
```

Summary of Annotated Parasitics:

+-----+				
Net Type	Count	Annotated (%)	Not Annotated (%)	
+-----+				
total	241	139 57.68%	102 42.32%	

Certus Text Command Reference
Signoff ECO Commands

+-----+-----+-----+-----+			
0-term:floating (*)	37	0 0.00%	37 100.00%
no drive (*)	32	0 0.00%	32 100.00%
1-term:no load	33	0 0.00%	33 100.00%
real net (complete)	139	139 100.00%	0 0.00%
real net (broken)	0	0 0.00%	0 0.00%
zero capacitance net	0	0 0.00%	0 0.00%
+-----+-----+-----+-----+			

Note for Net Types marked "(*)": Such nets are never timed, but reported here for informational purpose.

+-----+-----+-----+-----+			
Annotated	Res (KOhm)	Cap (pF)	XCap (pF)
+-----+-----+-----+-----+			
Count	1592	1658	272
Value	8.0015	0.5732	0.0827
+-----+-----+-----+-----+			

*info: output from client tdsp_core(2).

=====

```
#####
# report_annotated_parasitics: Thu Feb 20 01:36:20 2025
#   view: setup_func
#   -list_not_annotated
#####
#
# list syntax: type: net_name [nrCap Cap nrXCap XCap nrRes Res]
```

Certus Text Command Reference

Signoff ECO Commands

unit: pF, Ohm

Summary of Annotated Parasitics:

Net Type	Count	Annotated (%)	Not Annotated (%)
total	3547	2553 71.98%	994 28.02%
assign (*)	59	0 0.00%	59 100.00%
0-term:floating (*)	876	0 0.00%	876 100.00%
no drive (*)	56	0 0.00%	56 100.00%
1-term:no load	3	0 0.00%	3 100.00%
real net (complete)	2553	2553 100.00%	0 0.00%
real net (broken)	0	0 0.00%	0 0.00%
zero capacitance net	0	0 0.00%	0 0.00%

Note for Net Types marked "(*)": Such nets are never timed, but reported here for informational purpose.

Annotated	Res (KOhm)	Cap (pF)	XCap (pF)
Count	38329	38418	5288
Value	145.6864	5.9198	0.7150

- The following command queries `report_annotated_parasitics` from the selected partitions `{ptn_wrapper tdsp_core}`:

```
certus> distribute_certus_command {report_annotated_parasitics} -designs {ptn_wrapper tdsp_core}
```

```
*info: output from client ptn_wrapper(1).
```

```
=====
```

```
#####
```

```
# report_annotated_parasitics: Thu Feb 20 01:48:58 2025
```

```
# view: ptn_wrapper_1_setup_func
```

```
# -list_not_annotated
```

```
#####
```

```
#
```

```
# list syntax: type: net_name [nrCap Cap nrXCap XCap nrRes Res]
```

```
# unit: pF, Ohm
```

```
# Summary of Annotated Parasitics:
```

+-----+				
Net Type	Count	Annotated (%)		Not Annotated (%)
+-----+				
total	241	139	57.68%	102 42.32%
+-----+				
0-term:floating (*)	37	0	0.00%	37 100.00%
no drive (*)	32	0	0.00%	32 100.00%
1-term:no load	33	0	0.00%	33 100.00%
real net (complete)	139	139	100.00%	0 0.00%
real net (broken)	0	0	0.00%	0 0.00%
zero capacitance net	0	0	0.00%	0 0.00%
+-----+				

Note for Net Types marked "(*)": Such nets are never timed, but reported here for informational purpose.

+-----+				
Annotated	Res (KOhm)	Cap (pF)	XCap (pF)	
+-----+				
Count	1592	1658	272	
Value	8.0015	0.5732	0.0827	
+-----+				

- The following command queries `checkPlace` from each partition:

```
certus> distribute_certus_command {checkPlace}
```

```
*info: output from client dtmf_recvr_core(0).
```

```
=====
```

```
**WARN: (IMPSP-362): Site 'CoreSite' has one std.Cell height, so ignoring its X-symmetry.
```


Type 'man IMPSP-362' for more detail.

****Info:** (IMPSP-307): Design contains fractional 17 cells.

Estimated cell power/ground rail width = 0.213 um

Begin checking placement ... (start mem=2379.8M, init mem=2519.7M)

Region/Fence Violation: 1

*info: Placed = 766 (Fixed = 119)

*info: Unplaced = 0

Placement Density:1.36%(2262/165793)

Placement Density (including fixed std cells):1.50%(2500/166031)

Finished checkPlace (total: cpu=0:00:00.1, real=0:00:01.0; vio checks:
cpu=0:00:00.0, real=0:00:00.0; mem=2485.6M)

*info: output from client ptn_wrapper(1).

=====

****WARN:** (IMPSP-362): Site 'CoreSite' has one std.Cell height, so ignoring its X-symmetry.

Type 'man IMPSP-362' for more detail.

****Info:** (IMPSP-307): Design contains fractional 17 cells.

Estimated cell power/ground rail width = 0.213 um

Begin checking placement ... (start mem=2359.7M, init mem=2495.3M)

*info: Placed = 115 (Fixed = 2)

*info: Unplaced = 0

Placement Density:0.81%(202/24977)

Placement Density (including fixed std cells):0.81%(202/24977)

Finished checkPlace (total: cpu=0:00:00.1, real=0:00:01.0; vio checks:
cpu=0:00:00.0, real=0:00:00.0; mem=2461.3M)

```
*info: output from client tdsp_core(2).
```

```
=====
```

```
**WARN: (IMPSP-362): Site 'CoreSite' has one std.Cell height, so ignoring its X-  
symmetry.
```

```
Type 'man IMPSP-362' for more detail.
```

```
**Info: (IMPSP-307): Design contains fractional 17 cells.
```

```
Estimated cell power/ground rail width = 0.213 um
```

```
Begin checking placement ... (start mem=2364.4M, init mem=2502.0M)
```

```
*info: Placed = 2495
```

```
*info: Unplaced = 0
```

```
Placement Density:18.07%(7851/43437)
```

```
Placement Density (including fixed std cells):18.07%(7851/43437)
```

```
Finished checkPlace (total: cpu=0:00:00.2, real=0:00:01.0; vio checks:  
cpu=0:00:00.1, real=0:00:00.0; mem=2470.0M)
```

Related Commands

- `start_distributed_eco_clients`

export_distributed_eco

```
export_distributed_eco  
[-help]  
[-designs <string>]  
[-export_only innovus_dbs | sta_dbs] -out_dir <string>  
[-overwrite]
```

Exports Innovus DBs and Tempus STA sessions in `out_dir` at the end of the Certus-based optimization.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>export_distributed_eco</code> parameter.
<code>-designs</code> <code><string></code>	Specifies the list of modules for which an Innovus partition DB needs to be exported. <i>Default:</i> <code>all</code>
<code>-export_only</code> <code>innovus_dbs </code> <code>sta_dbs</code>	Exports client Tempus sessions (<code>sta_dbs</code>) or Innovus DBs (<code>innovus_dbs</code>).
<code>-out_dir</code> <code><string></code>	Specifies the name of the output directory in which all data will be written. This output directory includes <code>Innovus_DBs</code> and <code>STA_dbs</code> directories.
<code>-overwrite</code>	Overwrites the existing <code>out_dir</code> if the same directory is provided again.

Examples

The following command saves Innovus DBs for all partitions in the output directory. These saved Innovus DBs can be used for the block-level flows to clean up any remaining routing DRCs or to generate Verilog/DEF. However, these saved Innovus DBS should not be used to generate any timing reports or to perform further timing/power optimization, as timing cannot be checked further at block-level. This is because Certus has made netlist updates based on a timing picture from the sub-system/full-chip level:

```
certus> export_distributed_eco -export_only {innovus_dbs} -out_dir output
```

Related Commands

- `time_design_signoff`

get_design_mode

```
get_design_mode  
[-help]  
[-flowEffort]  
[-powerEffort]
```

Returns the current settings of the `set_design_mode` command.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>get_design_mode</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man get_design_mode</code> .
<code>parameter_names</code>	Displays information for the specified parameters. See the <code>set_design_mode</code> command for descriptions of the <code>set_design_mode</code> parameters that you can specify.

Examples

- The following command displays the current settings of `set_design_mode -flowEffort`:

```
certus > get_design_mode -flowEffort  
-flowEffort standard # enums={express standard extreme}, default=standard  
standard
```
- The following command displays the current settings of `set_design_mode -powerEffort`:

```
certus > get_design_mode -powerEffort  
-powerEffort none # enums={none low high}, default=none  
none
```

Related Commands

- `set_design_mode`

generate_summary

```
generate_summary  
[-help]  
[-logvFileName <string>]  
[-outputFile <string>]
```

Displays the Certus log summary report in a text file. The log summary report includes the following details:

- Runtime details of the full flow.
- Tool versions for Certus Manager, Certus Clients, and Tempus Clients.
- Active views for each Tempus Client.
- Partition Name, Netlist Size, the Number of Replications, and Log Pointer for each Certus Client.
- Number of CPUs enabled for Certus Manager, Certus Clients, and Tempus Clients.
- Size of the netlist on the Certus Manager, Certus Clients, and Tempus Clients.
- RC annotation summary for each partition loaded as a Certus client.
- QoR/TAT/Memory summary from `time_design_signoff`, ECO optimization, and `implement_distributed_eco` steps.
- Mean displacement, Max displacement, Instances moved, and Errors in `refinePlace` for each partition.
- Final Routing DRC after each `implement_distributed_eco` call for each partition.
- Initial density (from the first optimization step) and the final density (from the last optimization step) for each partition.
- List of error codes and DIAGs for Certus Manager, Tempus Clients, and Certus Clients.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>generate_summary</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man generate_summary.</code>
<code>-logvFileName</code> <string>	Specifies the name of the log file. By default, the log file name is <code>certus.logv</code>. Note: This parameter is optional.
<code>-outputFile</code> <string>	Specifies the name of the output file, which includes the log summary details for the specified log file. For the log file, <code>certus.logv</code>, the default log summary file name is <code>certus.logv.summary.txt</code>. Note: This parameter is optional.

Examples

- The following command generates the summary report, `certus.logv.summary.txt`, for the Certus log file, `certus.logv`:
- The sample script below performs distributed hold optimization in full-flat mode and ends with the ECO implementation step. This step performs ECO routing and parasitic extraction for each Certus client, followed by a timing update with updated parasitics for the timing clients.

```
certus> generate_summary

set eco_setup_view_list "V1 V2"
set eco_hold_view_list "V1 V2"
set eco_dynamic_view V1
set eco_leakage_view V2
set_multi_cpu_usage -localCpu 12 -remoteHost 2 -cpuPerRemoteHost 8
set_eco_opt_mode -distributed_opt_block_restore_file blkRestore.txt
distribute_read_design -restore_design {save_db_dmmmc_V1 save_db_dmmmc_V2} -outdir
paradime_Output
```

```
set_multi_cpu_usage -localCpu 12 -remoteHost 3 -cpuPerRemoteHost 8
start_distributed_eco_clients
set_eco_opt_mode -verbose true
time_design_signoff -outDir INITIAL_TDS -report_power
set_eco_opt_mode -eco_file_prefix ECO/drv
eco_opt_design -drv
set_eco_opt_mode -eco_file_prefix ECO/setup
eco_opt_design -setup
set_eco_opt_mode -eco_file_prefix ECO/hold
eco_opt_design -hold
implement_distributed_eco
time_design_signoff -outDir FINAL_TDS -report_power
generate_summary
```

The following is a sample of the resulting text report (certus.logv.summary.txt):

```
#####
#
#  Cadence Certus(TM) Solution Summary File
#  Created on Wednesday, Jun 25, 2025 22:25:56
#
#####
Run Started on:    Wed, Jun 25, 2025 22:21:04
Run Terminated on: Wed, Jun 25, 2025 22:25:55
Full Flow Runtime: 00:04:51

+-----+-----+
|      Process      |   Version   |
+-----+-----+
| Certus Manager Version | 25.11-e066_1 |
| Certus Client Version  | 25.11-e066_1 |
| Tempus Client Version  | 25.11-e056_1 |
+-----+-----+

+-----+-----+-----+
|Tempus Client(s)|           View(s)           |
|               | Log Pointer                 |
+-----+-----+-----+
| tempus_client0 |           V1                 |
|   paradime_Output/3946380_87d828f0-63c9-4eae-90de-7d66d3b450be_0.log   |
| tempus_client1 |           V2                 |
|   paradime_Output/3946380_87d828f0-63c9-4eae-90de-7d66d3b450be_1.log   |
```


Certus Text Command Reference

Signoff ECO Commands

Certus Client(s)	Partition Name	Netlist Size	Replicated Modules	
Log Pointer				
Certus_client0	dtmf_recvr_core	766	1	
DistOpt/3946380_0.log				
Certus_client1	ptn_wrapper	113	2	
DistOpt/3946380_1.log				
Certus_client2	tdsp_core	2495	1	
DistOpt/3946380_2.log				
Process	CPUs enabled			
certus_manager	12			
tempus_client0	8			
tempus_client1	8			
certus_client0	8			
certus_client1	8			
certus_client2	8			
Netlist Size	StdCell			
Tempus Client	3490			
Certus Manager	3490			

Block Annotated Parasitics Report 1

Certus Text Command Reference

Signoff ECO Commands

Block		RC Corner		View		Net Type	
Count	Annotated (%)	Not Annotated (%)					
dtmf_recvr_core		rcMax		V1		real net (complete)	
861	848 98.49%	13 1.51%					
ptn_wrapper		rcMax		ptn_wrapper_1_V1		real net (complete)	
139	139 100.00%	0 0.00%					
tdsp_core		rcMax		V1		real net (complete)	
2553	2553 100.00%	0 0.00%					

#####QoR
Summary#####

Legend: H: Hold; S: Signal Nets; C: Clock Nets

time_design_signoff reports summary for all Path Group

Step		WNS	TNS	VP	WNS (H)
TNS (H)	VP (H)	m_cap (S)	m_cap (C)	m_tran (S)	m_tran (C)
LkgPwr	IntPwr	SwPwr	TPwr	Runtime	Memory
Added	Resized	Deleted			
time_design_signoff 1		-2.455	-66.878	49	-4.364
-153.500	176	0 (0)	3 (3)	1 (8)	5 (43)
0.0005	0.3392	0.3813	0.721	0:00:16.0	2797.6M
eco_opt -drv Initial		-2.455	-66.879	49	-4.364
-153.500	176	0 (0)	3 (3)	4 (22)	5 (43)
eco_opt -drv Final		-2.455	-66.879	49	-4.364
-153.500	176	0 (0)	3 (3)	0 (0)	5 (43)
				0:01:11	3886.0M
					3

Certus Text Command Reference
Signoff ECO Commands

	1								
	eco_opt -setup Initial		-2.455		-66.879		49		-4.364
	-153.500		176		0 (0)		3 (3)		0 (0)
	eco_opt -setup Final		-2.426		-38.757		16		-4.364
	-153.498		176		0 (0)		3 (3)		0 (0)
							0:00:14.0		3861.2M
									3
	27								
	eco_opt -hold Initial		-2.426		-38.757		16		-4.364
	-153.498		176		0 (0)		3 (3)		0 (0)
	eco_opt -hold Final		-2.426		-38.755		16		0.002
	0.000		0		0 (0)		3 (3)		0 (0)
							0:00:42.0		3950.9M
									218
	ecoRoute								
							0:00:16.0		
	extractRC_time								
							0:00:20.0		
	tps_client_update								
							0:00:07.0		
	time_design_signoff 2		-2.282		-36.935		27		-0.192
	-1.500		24		0 (0)		3 (3)		0 (0)
	0.0006		0.4457		0.3591		0.8054		0:00:14.0
									3653.4M
<pre> +-----+-----+-----+-----+-----+ +-----+-----+-----+-----+-----+ +-----+-----+-----+-----+-----+ +-----+-----+-----+-----+-----+ </pre>									
##### Time Design									
Signoff Summary #####									
<pre> +-----+-----+-----+-----+-----+ +-----+-----+-----+-----+-----+ </pre>									
	Step		Path Group		WNS		TNS		

Certus Text Command Reference
Signoff ECO Commands

VP	WNS (H)	TNS (H)	VP (H)
time_design_signoff1	all	-2.455	-66.878
49	-4.364	-153.500	176
time_design_signoff1	reg2reg	-1.033	-27.878
33	-3.170	-69.129	37
time_design_signoff1	in2reg	1.744	0.000
0	-4.364	-120.048	153
time_design_signoff1	reg2out	-2.455	-39.001
16	0.413	0.000	0
time_design_signoff1	in2out	1.420	0.000
0	0.590	0.000	0
time_design_signoff2	all	-2.282	-36.935
27	-0.192	-1.500	24
time_design_signoff2	reg2reg	-0.090	-0.482
11	-0.192	-0.992	13
time_design_signoff2	in2reg	0.804	0.000
0	-0.057	-0.508	11
time_design_signoff2	reg2out	-2.282	-36.452
16	0.416	0.000	0
time_design_signoff2	in2out	0.308	0.000
0	0.519	0.000	0

DRV Opt
#####

DRV Optimization

Attribute	Value
Timing Mode	SOCV PBA
Analysis Mode	WFP SI
EOSDB	On
max_tran fixing rate nets (terms)	100.0%(100.0%)
max_cap fixing rate nets (terms)	N/A%(N/A%)

Certus Text Command Reference

Signoff ECO Commands

Refine Place Movements	Mean displacement	Max displacement	Instances Moved
ERROR ID			
Certus_Client0	0.00 um	0.00 um	N/A
N/A			
Certus_Client1	N/A	N/A	N/A
N/A			
Certus_Client2	0.00 um	0.00 um	N/A
N/A			

Failure reasons for drv:

Could not be fixed (Clock Nets)	Number of nets
ClockNet	5

Setup Opt
#####

Setup Optimization

Attribute	Value
Timing Mode	SOCV PBA
Analysis Mode	WFP SI
EOSDB	On
WNS fixing rate	1.2%
TNS fixing rate	42.0%
VP fixing rate	67.3%

Refine Place Movements	Mean displacement	Max displacement	Instances Moved
ERROR ID			
Certus_Client0	0.00 um	0.00 um	N/A
N/A			

Certus Text Command Reference

Signoff ECO Commands

Certus_Client1	0.00 um	0.00 um	N/A
N/A			
Certus_Client2	0.00 um	0.00 um	N/A
N/A			

```

+-----+-----+-----+-----+
+-----+

```

Failure reasons for setup:

Could not be fixed (Data Nets)	Number of nets
SetupNotImprovingBuff	23
HoldWNSDegradationRsz	2
SetupNotImprovingDel	1

Hold Opt
#####

Hold Optimization

Attribute	Value
Timing Mode	SOCV PBA
Analysis Mode	WFP SI
EOSDB	On
WNS fixing rate	100.0%
TNS fixing rate	100.0%
VP fixing rate	100.0%

```

+-----+-----+-----+-----+
+-----+

```

Refine Place Movements	Mean displacement	Max displacement	Instances Moved
ERROR ID			

```

+-----+-----+-----+-----+
+-----+

```

Certus_Client0	0.00 um	0.00 um	N/A
N/A			
Certus_Client1	N/A	N/A	N/A
N/A			
Certus_Client2	0.00 um	0.00 um	N/A
N/A			

Certus Text Command Reference
Signoff ECO Commands

```
+-----+-----+-----+-----+
+-----+
```

```
#####Final Routing
DRC#####
```

```
+-----+-----+
|Certus Client(s)|  Final DRCvio call#1  |
+-----+-----+
| certus client0 |           0           |
| certus client1 |           0           |
| certus client2 |           0           |
+-----+-----+
```

```
#####Change in
Density#####
```

```
+-----+-----+-----+
|Certus Client(s)| Initial Density |  Final Density  |
+-----+-----+-----+
|certus client0 |    1.365%    |    2.284%    |
|certus client1 |    0.812%    |    0.812%    |
|certus client2 |   18.075%   |   18.662%   |
+-----+-----+-----+
```

```
#####DIAGs & ERROR
ID#####
```

```
+-----+-----+-----+-----+
-----+
-----+
|      Process      |
|      DIAGs       |
|      |           |
+-----+-----+-----+-----+
-----+
-----+
| Certus manager   | N/A
|
| certus_client0   | N/A
|
| certus_client1   | N/A
```

```

|
| certus_client2 | N/A

```

```

|
| tempus_client0 | N/A

```

```

|
| tempus_client1 | N/A

```

```

|
+-----+-----+
-----+
-----+
+-----+-----+
|      Process      |      ERROR ID      |
+-----+-----+
| Certus manager    | IMPEXT-7082         |
| certus_client0    | IMPESO-301          |
| certus_client1    | IMPESO-301          |
| certus_client2    | IMPESO-301          |
| tempus_client0    | IMPEXT-7082         |
| tempus_client1    | IMPEXT-7082         |
+-----+-----+

```

- The following command generates the summary report, `certus.logv1.summary.txt`, for the specified log file, `certus.logv1`:

```

certus> generate_summary -logvFileName certus.logv1
*info: Generating summary report from log:  ['/workspace/data/certus.logv1']
*info: summary report generated in  certus.logv1.summary.txt

```

Certus summary report: `certus.logv1.summary.txt`

- The following command generates the output file, `out.txt`, which includes the log summary details for the specified log file, `certus.logv1`.

```

certus> generate_summary -logvFileName certus.logv1 -outputFile out.txt
*info: Generating summary report from log:  ['/workspace/data/certus.logv1']
*info: summary report generated in  out.txt

```

Certus summary report: `out.txt`

Related Commands

- `eco_opt_design`
- `start_distributed_eco_clients`
- `time_design_signoff`

implement_distributed_eco

```
implement_distributed_eco  
[-help]  
[-continue]  
[-force]
```

Performs ECO routing, parasitic extraction, and parasitic back-annotation on the Certus client manager as well as the Tempus clients, followed by a timing update. When you run this command, the Certus client manager session is ready for any signoff timing analysis post ECO implementation, and you can start the next session of ECO closure.

Parameters

-help	Outputs a brief description that includes type and default information for each <code>implement_distributed_eco</code> parameter.
-continue	Allows the <code>implement_distributed_eco</code> command to continue where the previous call of that command failed. This can be useful in scenarios where parasitic extraction failed due to some missing data. Using this parameter, you can then update the current session and restart the <code>implement_distributed_eco</code> command to perform only parasitic extraction.
-force	Performs ECO routing and parasitic extraction for all blocks (except those not loaded on a Certus client). By default, the <code>implement_distributed_eco</code> command performs ECO routing and parasitic extraction only on the blocks where ECO changes have been made.

Examples

- The following command runs incremental routing and parasitic extraction on each Certus client that was modified by the ECO optimization step, and back-annotates the parasitics on the Tempus timing sessions:

```
certus> implement_distributed_eco
```

- In case `implement_distributed_eco` errors out during parasitic extraction, you can update a few settings on the Certus clients and restart from the `extractRC` step using the following commands:

```
certus> set_eco_opt_mode -distributed_opt_pre_extraction_tcl pre_extractRC.tcl  
certus> implement_distributed_eco -continue
```

Related Commands

- `start_distributed_eco_clients`

set_eco_opt_mode

```
set_eco_opt_mode
[-help]
[-custom_distribution_file file]
[-distributed_opt_block_restore_file file]
[-distributed_opt_max_clients_per_host value]
[-distributed_opt_out_dir directoryname]
[-distributed_opt_pre_eco_route_tcl file]
[-distributed_opt_post_eco_route_tcl file]
[-distributed_opt_pre_extraction_tcl file]
[-distributed_opt_post_extraction_tcl file]
[-distributed_opt_post_restore_design_tcl file]
[-distributed_opt_pre_restore_design_tcl file]
```

Sets the global parameters for ECO. The parameters that you specify with the `set_eco_opt_mode` command are then used automatically when you run ECO optimization by running the `eco_opt_design` command.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>set_eco_opt_mode</code> parameter.
<code>-custom_distribution_file file</code>	Specifies a file that provides differentiated resource distribution information for CPU and memory configurations of Certus and Tempus clients. This parameter supports various resource requirements, helping ensure optimal performance. Note: This parameter must be specified before <code>distribute_read_design</code>. <i>Default:</i> "" For details, see the Recommended Guidelines for Certus chapter in the <i>Certus User Guide</i>
<code>-distributed_opt_block_restore_file file</code>	

`-distributed_opt_block_restore_file file`

Provides a list of pointers to the Innovus database that can be loaded on Certus clients when calling the `start_distributed_eco_clients` command. If some partitions in the Certus client manager netlist are not included in the block restore file, those partitions will be considered don't touch during ECO closure.

If the Certus client manager has a reduced database when running in High Capacity ECO mode, partitions not present in the Certus client manager will not be loaded by any Certus client.

Example:

`blockName1 INVS_DB_PATH_for_block1`

`blockName2 INVS_DB_PATH_for_block2`

and so on

Any partition that is listed as part of –

`dont_touch_partition_list` or –

`touch_partition_list` or –

`intra_partition` or –

`inter_partition` or –

`inter_partition_exclusive` must be specified in the block restore file; however, for a few partitions that are excluded from ECO, only the partition name needs to be specified.

Example:

`blockName3` is a nested partition inside `blockName2`. If Signoff ECO optimization is not allowed inside `blockName3`, and –

`intra_partition` partition fixing is applied to `blockName2` only, then you must specify the partition name, `blockName3`, in the block restore file, and skip specifying its

	Innovus database: <pre>blockName1 INVS_DB_PATH_for_block1 blockName2 INVS_DB_PATH_for_block2 blockName3</pre>
<code>-distributed_opt_max_clients_per_host <value></code>	When a value greater than 0 is provided, the Certus client manager is instructed to reuse the Tempus client's machines when starting Certus clients using the <code>start_distributed_eco_clients</code> command. This allows sharing CPUs between Tempus and Certus clients. Note that memory usage on those machines is equal to the sum of one of the Tempus client process memory and N Certus client process memory, where N is the value set by this parameter. <i>Default: 0</i>
<code>-distributed_opt_out_dir directoryname</code>	Specifies the directory name where the Certus clients' log files will be generated. By default, the tool uses <code>DistOpt/</code> in the current directory.
<code>-distributed_opt_pre_eco_route_tcl file</code>	Provides a TCL script that each Certus client will source before performing ECO routing during the <code>implement_distributed_eco</code> command.
<code>-distributed_opt_post_eco_route_tcl file</code>	Provides a TCL script that each Certus client will source after performing ECO routing during the <code>implement_distributed_eco</code> command.
<code>-distributed_opt_pre_extraction_tcl file</code>	Provides a TCL script that each Certus client will source before performing parasitic extraction during the <code>implement_distributed_eco</code> command.

<code>-distributed_opt_post_extraction_tcl file</code>	Provides a TCL script that each Certus client will source after performing parasitic extraction during the <code>implement_distributed_eco</code> command.
<code>-distributed_opt_post_restore_design_tcl file</code>	Provides a TCL script that each Certus client will source after loading the Innovus DB. <i>Default: ""</i>
<code>-distributed_opt_pre_restore_design_tcl file</code>	Provides a TCL script that each Certus client will source before loading the Innovus DB. <i>Default: ""</i>

Note: The above `set_eco_opt_mode` parameters are specific to Certus only. For details related to other parameters of the [set_eco_opt_mode](#) command, refer to the *Tempus Text Command Reference* document.

Examples

- The following command refers to a file (`blocks.txt`) pointing to all Innovus DBs that are opened for ECO:

```
set_eco_opt_mode -distributed_opt_block_restore_file blocks.txt
```

Below is a template for the `blocks.txt` file:

```
blockName1 <dirName>/blockName1.enc.dat
```

```
blockName2 <dirName>/blockName2.enc.dat
```

Refer to the [Template Scripts](#) section of the *Certus User Guide*.

- The following command refers to a file (`distribution.tcl`) referring to each client requesting a separate machine:

```
set_eco_opt_mode -custom_distribution_file distribution.tcl
```

where (`distribution.tcl`) file includes:

```
module=topChip nCpu=16 {bsub -q batch -W 100:00 -P SSV:26.1:Certus -o /dev/null -n
8 -R 'select\[OSREL==EE80\] rusage\[mem=250000,tmp=50000\] span\[hosts=1\]'} }
```


set_design_mode

```
set_design_mode  
[-help]  
[-reset]  
[-flowEffort]  
[-powerEffort]
```

This command, when used in combination with `eco_opt_design -all`, enables you to choose between the timing closure flow or the timing and power closure flow in Certus cockpit, where `eco_opt_design -all` honors the signoff timing from the Tempus clients and performs the physical-aware distributed optimization through each Certus client.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>set_design_mode</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man set_design_mode</code> .
<code>-flowEffort</code> <code>{express standard extreme}</code>	Guides the effort of the main command (<code>eco_opt_design -all</code>) for timing closure. • <code>express</code>: In the Certus cockpit, this flow runs similarly to the standard flow. • <code>standard</code>: Configures the flow to improve the quality of results and the full-flow turnaround time. This flow is useful for the majority of the designs. • <code>extreme</code>: Configures the flow to produce higher accuracy results, reducing the turnaround time (TAT) significantly. This flow is useful for designs where timing closure is challenging. <i>Default:</i> <code>standard</code>

<code>-powerEffort</code>	Specifies the effort to reduce the dynamic and leakage power. The tradeoff between dynamic and leakage power can also be controlled. • <code>none</code>: No power optimization is performed, and power is not considered during timing optimization. • <code>low</code>: Power is considered during timing optimization, and power-aware optimization is performed in all flow stages. • <code>high</code>: Additional algorithms are used to minimize the power consumption. The turnaround time with this option will be longer than using <code>low</code> or <code>none</code>. <i>Default:</i> <code>none</code>
<code>-reset</code>	Resets parameters to their default values. The <code>-reset</code> parameter must be the first parameter specified. If you specify <code>-reset</code> by itself, the software resets all <code>set_design_mode</code> parameters to their default values. Any values that do not already match the default value and are reset will be reported.

Note: The above `set_design_mode` parameters are specific to Certus only. For details related to other parameters of the [set_design_mode](#) command, refer to the *Tempus Text Command Reference* document.

Examples

- The following command performs Timing closure (DRV, SETUP, HOLD) in PBA mode:

```
certus> set_design_mode -powerEffort none -flowEffort standard
certus> eco_opt_design -all
```

- The following command performs Timing and Power Closure (DRV, SETUP, HOLD, and POWER) in PBA mode:

```
certus> set_design_mode -powerEffort low/high -flowEffort standard
certus> eco_opt_design -all
```

- The following command performs Timing Closure with aggressive Setup fixing at the expense of Hold timing in PBA mode:

```
certus> set_design_mode -powerEffort none -flowEffort extreme
certus> eco_opt_design -all
```

Related Commands

- [set_eco_opt_mode](#)
- [get_design_mode](#)

start_distributed_eco_clients

```
start_distributed_eco_clients  
[-help]
```

Launches the Certus clients, which will each load the respective Innovus database. The tool depends on the input file specified with `set_eco_opt_mode -distributed_opt_block_restore_file` to obtain pointers to these Innovus databases. If some partitions in the Certus client manager netlist are not part of the block restore file, those partitions are considered don't touch during ECO closure.

When the Certus client manager has a reduced database while running in High Capacity ECO mode, partitions not present in the Certus client manager will not be loaded by any Certus client. Certus clients can either collect their own machines or reuse Tempus clients' machines when specifying `set_eco_opt_mode -distributed_opt_max_clients_per_host` with a value greater than 0. The Certus clients are instructed by `set_multi_cpu_usage -cpuPerRemoteHost` to identify how many threads can be used by the clients. In case the Certus clients have to collect more machines from the farm, you have to specify the correct farm requests using the `set_distribute_host` command.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>start_distributed_eco_clients</code> parameter.
--------------------	---

Related Commands

- `implement_distributed_eco`

time_design_signoff

```
time_design_signoff
[-help]
[-no_expanded_views]
[-report_dir <string>]
[-report_prefix <string>]
[-report_full_clock_path]
[-report_power]
```

Generates signoff timing reports for each view from the Tempus clients. The command prints out a timing summary table per view and all the views combined.

Parameters

-help	Outputs a brief description that includes type and default information for each <code>time_design_signoff</code> parameter.
-no_expanded_views	Disables printing of expanded timing per view.
-report_dir <i>string</i>	Specifies the directory to save reports.
-report_prefix <i>string</i>	Specifies the prefix for timing report file names.
-report_full_clock_path	When this parameter is specified, the detailed timing report generated includes the full clock path.
-report_power	When this parameter is specified, <code>time_design_signoff</code> prints a detailed power summary report from the power view.

Examples

- The following command prints the Setup/Hold/DRV timing summary:

```
certus> time_design_signoff
```

```
-----
                        time_design_signoff Summary
-----
```

Certus Text Command Reference
Signoff ECO Commands

Setup mode	all	reg2reg	in2reg	reg2out	in2out
WNS (ns):	-1.797	-0.536	-1.797	-0.244	N/A
TNS (ns):	-2054.234	-266.725	-1798.030	-0.244	N/A
Violating Paths:	8735	3402	5402	1	0
func-rcMax	-1.797	-0.536	-1.797	-0.244	N/A
	-2054.234	-266.725	-1798.030	-0.244	N/A
	8735	3402	5402	1	0

DRVs	Signal nets		Clock nets	
	Nr nets(terms)	Worst Vio	Nr nets(terms)	Worst Vio
max_cap	16 (16)	-0.346	0 (0)	0
max_tran	442 (690)	-2.117	0 (0)	0
max_fanout	0 (0)	0	0 (0)	0

Hold mode	all	reg2reg	in2reg	reg2out	in2out
WNS (ns):	-0.362	-0.362	0.143	-0.294	N/A
TNS (ns):	-340.889	-256.428	0.000	-84.461	N/A
Violating Paths:	1773	1262	0	511	0
func-rcMin	-0.362	-0.362	0.143	-0.294	N/A
	-340.889	-256.428	0.000	-84.461	N/A
	1773	1262	0	511	0

- The following command reports power summary from the power view:

```
certus> time_design_signoff -report_power
```

time_design_signoff Summary					
Setup mode	all	reg2reg	in2reg	reg2out	in2out

Certus Text Command Reference
Signoff ECO Commands

+-----+-----+-----+-----+-----+-----+						
	WNS (ns):		-1.797		-0.536	
	TNS (ns):		-2054.234		-266.725	
	Violating Paths:		8735		3402	
+-----+-----+-----+-----+-----+-----+						
	funct-rcMax		-1.797		-0.536	
			-2054.234		-266.725	
			8735		3402	
+-----+-----+-----+-----+-----+-----+						
+-----+-----+-----+-----+-----+-----+						

+-----+-----+-----+-----+-----+-----+																			
					Signal nets						Clock nets								
	DRVs	+-----+-----+-----+-----+-----+-----+																	
					Nr nets(terms)			Worst Vio						Nr nets(terms)			Worst Vio		
+-----+-----+-----+-----+-----+-----+																			
	max_cap		16 (16)			-0.346			0 (0)			0							
	max_tran		442 (690)			-2.117			0 (0)			0							
	max_fanout		0 (0)			0			0 (0)			0							
+-----+-----+-----+-----+-----+-----+																			

+-----+-----+-----+-----+-----+-----+						
	Hold mode		all		reg2reg	
+-----+-----+-----+-----+-----+-----+						
	WNS (ns):		-0.362		-0.362	
	TNS (ns):		-340.889		-256.428	
	Violating Paths:		1773		1262	
+-----+-----+-----+-----+-----+-----+						
	funct-rcMin		-0.362		-0.362	
			-340.889		-256.428	
			1773		1262	
+-----+-----+-----+-----+-----+-----+						

Power report summary using power view : funct-rcMax

Total Power

```
-----
-----
Total Internal Power:      1130.32884617      62.2254%
Total Switching Power:    667.26277622      36.7333%
Total Leakage Power:      18.91418979       1.0412%
Total Power:              1816.50581218
```

```
-----
-----
Group                      Internal  Switching  Leakage    Total
Percentage
```

Certus Text Command Reference
Signoff ECO Commands

	Power	Power	Power	Power
(%)				

Sequential	675.1	95.71	4.636	775.5
42.69				
Macro	123.4	6.658	3.46	133.5
7.349				
IO	0	0	0	
0				
0				
Combinational	284	491.3	10.39	785.7
43.26				
Clock (Combinational)	24.94	67.43	0.2621	92.63
5.099				
Clock (Sequential)	22.83	6.155	0.1692	29.16
1.605				

Total	1130	667.3	18.91	1817
100				

Related Commands

- `export_distributed_eco`

System Commands - Basic Tool-Tcl Commands

start_gui

```
start_gui  
[-help]  
[-designs <string>]
```

Displays the Certus Manager GUI.

The `start_gui -designs` command launches the GUI of the Certus Clients.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>start_gui</code> parameter. For a detailed description of the command and all its parameters, use the <code>man</code> command: <code>man start_gui</code>.
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-designs <string>	Specifies the list of partitions for which <code>start_gui</code> is executed. Note: The <code>start_gui -designs</code> command can be run at any point after <code>start_distributed_eco_clients</code> is specified. The tool waits for the design to be loaded on the Certus Client(s) before launching the GUI. The Certus Manager shell remains inaccessible until all Certus Clients GUIs are closed. This GUI should be used solely as a viewer to display physical constraints, such as placement blockage, routing blockage, routing congestion, local density, and the availability of legal locations. The following warning is displayed when you run <code>start_gui -designs</code> after <code>start_distributed_eco_clients</code>: Operations done through the Certus Clients GUI should not impact routing, should not change <code>dont_touch/dont_use</code> attribute, should not trigger a new Parasitic extraction or a reset of parasitic, should not perform any timing report or timing query. Purpose of this GUI is to view the layout, query any physical attribute/property from the Certus Clients or change some instances placement status, and anything else is prohibited as it can desynchronize the Certus client manager from the Certus clients.
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Examples

- The following command launches the GUI for the Certus clients for the specified partitions, `ptn_wrapper` and `tdsp_core`:

```
certus> start_gui -designs {ptn_wrapper tdsp_core}
```
- The following command launches the GUI for all the partitions loaded as Certus clients:

```
certus> start_gui -designs all
```

Related Commands

- `start_distributed_eco_clients`

